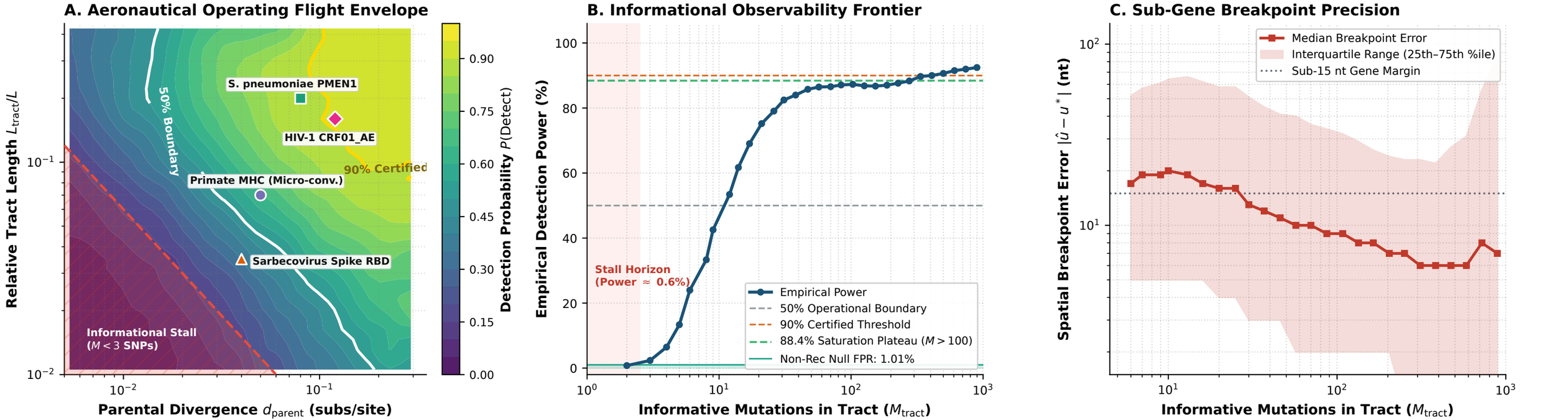
A. Aeronautical Operating Flight Envelope
S. pneumoniae PMEN1
HIV-1 CRF01_AE
90% Certified
50% Boundary
Primate MHC (Micro-conv.)
Sarbecovirus Spike RBD
Informational Stall
(M < 3 SNPs)
Relative Tract Length L_tract/L
Parental Divergence d_parent (subs/site)
Detection Probability P(Detect)
B. Informational Observability Frontier
Empirical Detection Power (%)
Stall Horizon
(Power ≈ 0.6%)
Empirical Power
50% Operational Boundary
90% Certified Threshold
88.4% Saturation Plateau (M > 100)
Non-Rec Null FPR: 1.01%
Informative Mutations in Tract (M_tract)
C. Sub-Gene Breakpoint Precision
Spatial Breakpoint Error |û − u*| (nt)
Median Breakpoint Error
Interquartile Range (25th–75th %ile)
Sub-15 nt Gene Margin
Informative Mutations in Tract (M_tract)